

SobolEngine#

<https://pytorch.org/docs/stable/generated/torch.quasirandom.SobolEngine.html>

Description:

The `torch.quasirandom.SobolEngine` is an engine for generating (scrambled) Sobol sequences. Sobol sequences are an example of low discrepancy quasi-random sequences. This implementation of an engine for Sobol sequences is capable of sampling sequences up to a maximum dimension of 21201. It uses direction numbers from <https://web.maths.unsw.edu.au/~fkuo/sobol/> obtained using the search criterion $D(6)$ up to the dimension 21201. This is the recommended choice by the authors. Art B. Owen. Scrambling Sobol and Niederreiter-Xing points. *Journal of Complexity*, 14(4):466-489, December 1998. I. M. Sobol. The distribution of points in a cube and the accurate evaluation of integrals. *Zh. Vychisl. Mat. i Mat. Phys.*, 7:784-802, 1967.

Function Signature

```
class torch.quasirandom.SobolEngine(dimension,  
scramble=False, seed=None)[source]#
```

Parameters

```
draw(n=1, out=None, dtype=None)[source]#
```

Parameters

Return type

Returns:

Tensor

Code Examples

```
>>> soboleng = torch.quasirandom.SobolEngine(dimension=5)  
>>> soboleng.draw(3)  
tensor([[0.0000, 0.0000, 0.0000, 0.0000, 0.0000],  
        [0.5000, 0.5000, 0.5000, 0.5000, 0.5000],  
        [0.7500, 0.2500, 0.2500, 0.2500, 0.7500]])
```

Detailed Documentation

Documentation

The `torch.quasirandom.SobolEngine` is an engine for generating (scrambled) Sobol sequences. Sobol sequences are an example of low discrepancy quasi-random sequences.

This implementation of an engine for Sobol sequences is capable of sampling sequences up to a maximum dimension of 21201. It uses direction numbers from <https://web.maths.unsw.edu.au/~fkuo/sobol/> obtained using the search criterion D(6) up to the dimension 21201. This is the recommended choice by the authors.

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`dimension (Int)` – The dimensionality of the sequence to be drawn

`scramble (bool, optional)` – Setting this to `True` will produce scrambled Sobol sequences. Scrambling is capable of producing better Sobol sequences. Default: `False`.

`seed (Int, optional)` – This is the seed for the scrambling. The seed of the random number generator is set to this, if specified. Otherwise, it uses a random seed. Default: `None`

Function to draw a sequence of `n` points from a Sobol sequence. Note that the samples are dependent on the previous samples. The size of the result is `(n,dimension)(n,dimension)(n,dimension)`.

`n (Int, optional)` – The length of sequence of points to draw. Default: 1

out (Tensor, optional) – The output tensor

dtype (torch.dtype, optional) – the desired data type of the returned tensor. Default: None

Function to draw a sequence of 2^m points from a Sobol sequence. Note that the samples are dependent on the previous samples. The size of the result is $(2^m, \text{dimension})$.

m (Int) – The (base2) exponent of the number of points to draw.

out (Tensor, optional) – The output tensor

dtype (torch.dtype, optional) – the desired data type of the returned tensor. Default: None

Function to fast-forward the state of the SobolEngine by n steps. This is equivalent to drawing n samples without using the samples.

n (Int) – The number of steps to fast-forward by.

Function to reset the SobolEngine to base state.